HSU, Ke-Cheng (Clio)

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Professional Summary

PhD candidate in Economics at HKUST specializing in empirical industrial organization and digital economics. Experienced working with large proprietary datasets, including e-commerce platform transactions and retail point-of-sale records. Builds reproducible, testable structural-estimation pipelines on HPC/SLURM and AWS, and communicates results with clear data visualizations. Experience estimating demand and pricing models and delivering counterfactual welfare and policy analysis for platform and retail markets, including merger simulation and market entry/exit counterfactuals conducted in PhD-level industrial organization coursework. Seeking an economics consulting or research economist role.

Education

The Hong Kong University of Science and Technology, Hong Kong

Doctor of Philosophy in Economics — Expected Graduation: June 2027

Research interests: Empirical industrial organization and digital economics.

Master of Philosophy in Economics — 2020–2022

CGA: 4.1/4.3.

Bachelor of Science in Mathematics and Economics, Minor in Actuarial Mathematics — 2015–2019

CGA: 3.9/4.3.

Awards: Dean's List (2016–2019); University Scholarship (730K HKD); Academic Achievement Medal; Postgraduate studentship (1.5M HKD); Redbird scholarship (40K HKD).

Experience

Research Assistant, National Taiwan University — 2020

- Modeled crisis intervention as regulator–creditor negotiation; derived and simulated equilibrium predictions to support policy design.

Teaching Assistant & Lecturer, Department of Economics, HKUST — 2020–present

- Lecturer for MSc Math Camp (lecturer evaluation 4.53/5); TA for PhD microeconomic theory and undergraduate information economics and macroeconomics (TA evaluation average 4.72/5).

Projects

Return Policy Design in E-Commerce *(Job Market Paper)*

- Built and estimated a structural model of purchase, return-shipping insurance take-up, and return decisions under monopoly pricing, using within-retailer switches between automatic and optional return-shipping insurance on Taobao (2017–2023; ∼62M RMB revenue; six apparel categories; ∼20–30% returns).
- Developed a reproducible, testable estimation pipeline on AWS and HPC/SLURM: version-controlled steps that clean and link orders, prices, insurance, and returns; scraped and OCR-digitized product attributes in Python; accelerated Monte Carlo simulation and constrained optimization in R/C++, with unit-tested code and checks that the estimator recovers true parameters on simulated data.
- Estimated category-specific demand, consumer uncertainty about product fit, and return responses using method of moments (GMM); used bootstrap inference to rank policies by seller profit, consumer surplus, and social welfare.
- Produced figures (purchase/return rates, insurance take-up, welfare shares by category) for slides and tables; counterfactuals show a discounted-price + restocking-fee menu beats automatic and optional insurance in most categories (∼75–78% of first-best welfare), with moderate fees (∼10–15%) when one rule must apply.

Inventory and Pricing under Resale Price Maintenance (RPM) *(w/ K. Kawaguchi, HKUST; K. Kawai, U. Tokyo)*

- Specified a dynamic oligopoly model of pricing and inventory under demand uncertainty and RPM, with equilibrium stockouts and unsold returns (35–40% of Japanese books printed are discarded), to quantify pro-competitive inventory effects versus waste from price commitment.
- Built title-by-market-by-month analysis datasets from monthly bookstore point-of-sale data (deliveries, sales, returns; 1,000+ stores) through a tested, end-to-end pipeline: linked products across sources via standardized ISBNs; drew representative samples across book categories; used Hausman-style price instruments to address price endogeneity; collected local demographic data to capture municipality-level market heterogeneity in the estimation; adjusted for launch timing using Amazon release dates.
- Tackled endogeneity in demand and supply regressions with instrumental variables (2SLS) and structural model inversion; validated the estimator on simulated data (Monte Carlo); applied LASSO and panel fixed effects to select demographic controls; implemented iterative simulation of equilibrium in C++.
- Showed that the classical static model overestimates marginal cost by ∼20%, quantifying the bias from ignoring dynamic demand uncertainty and inventory decisions.
- Produced descriptive statistics and weekly short-turnaround empirical figures (sales/return/stockout rates, demand distributions by category, substitution patterns) and counterfactuals on deposit-refund policies that cut unsold stock while preserving inventory incentives.

Skills

Languages: Mandarin (Native), English (Fluent), Cantonese (Basic).

Technical: Python, R, C++, LaTeX, AWS, HPC/SLURM; MLE, GMM, minimum distance estimation, EM algorithm, bootstrap inference, counterfactual welfare (profit) analysis.